

Software Requirements Specification (SRS) for Exam Registration System

Introduction

Purpose

The Exam Registration System is developed to automate and streamline the process of registering for exams in an educational institution. The system will allow students to register for exams efficiently, ensure administrators can manage registrations and schedules, and provide automated notifications. This will reduce paperwork, minimize manual errors, and enhance overall efficiency.

Document Conventions

This document follows IEEE SRS standards.

Functional and non-functional requirements are clearly categorized.

The system specifications use standard terminologies relevant to software engineering

Intended Audience and Reading Suggestions

This document is intended for:

Developers: Responsible for coding and implementing the system.

University Administrators: Managing the exam registration process.

IT Support Team: Maintaining and troubleshooting the system.

QA/Testers: Ensuring the system meets requirements.

End Users: Students who will register for exams.

Product Scope

The Exam Registration System is a web-based application that integrates with the university's student database.

It allows students to:

Register for exams based on their enrolled courses.

Pay exam fees (if applicable).

View their exam schedule.

Receive notifications and reminders

Administrators can:

Manage exam registrations.

Approve or reject applications.

Generate reports on exam registrations and trends.

References

University policies on exam registration.

IEEE 830-1998: Software Requirements Specification standard.

User manuals of similar existing systems

Overall Description

Product Perspective

The Exam Registration System is an independent web application that will be integrated that will be the university existing academic management system. It replaces manual registration processes, ensuring security, accuracy and user-friendliness.

Product Functions

The system will have the following features:

User Authentication: Secure login for students and administrators.

Exam Registration: Students can register for exams within specified deadlines.

Payment Integration (if required): Secure payment gateway for exam fees.

Schedule Management: Students can check their exam schedules, and administrators can modify them.

Notification System: Automated email/SMS alerts for registration confirmations, deadlines, and schedule updates.

Report Generation: Administrators can generate reports on student registrations and exam trends

User Characteristics

Students: Basic computer literacy, ability to navigate the system.

Administrators: Knowledge of academic policies and system management.

IT Support: Understanding of web application maintenance and security.

Constraints

Internet access is required for system functionality.

Secure authentication must be in place to protect user data.

The system must be scalable to support thousands of students.

Compliance with university and government data protection laws.

Assumptions and Dependencies

Users will access the system through a web browser.

The university's database is maintained and up to date.

Third-party services (email, SMS) are available for notifications

Specific Requirements

Functional Requirements

User Authentication & Management

Users must log in using a username and password.

Two-factor authentication for administrators.

Role-based access control for different user types.

Exam Registration

Students can view available exams based on their course.

Register for exams within a specified timeframe.

Receive a confirmation email upon successful registration.

Payment Processing (if applicable) - Secure payment gateway integration.

Automatic receipt generation for payments.

Exam Schedule Management

Students can view and download exam schedules.

Administrators can modify schedules and notify students.

Notifications and Alerts

Email/SMS reminders for registration deadlines and schedule changes.

Alerts for unsuccessful payments or pending approvals.

Administrative Control Panel

Dashboard for viewing and managing registrations.

Generate reports on exam registration data.

Export data for administrative use.

Non-Functional Requirements

Security: Data encryption, secure authentication, and access control.

Usability: Intuitive UI design for easy navigation.

Performance: Capable of handling thousands of users simultaneously.

Availability: 99.9% uptime guarantee.

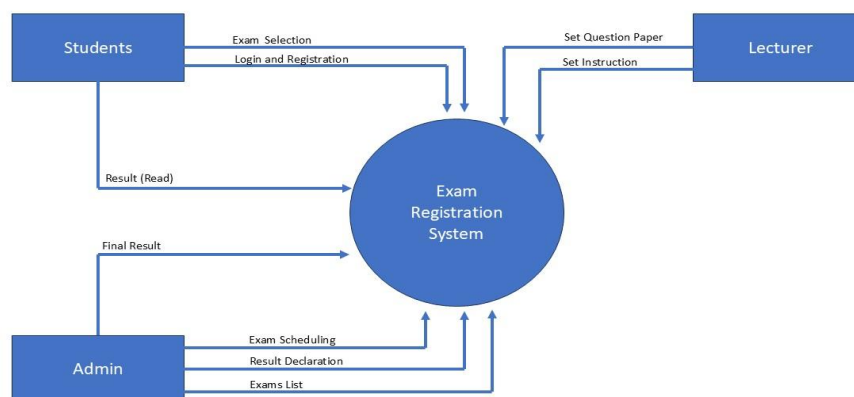
Scalability: Should accommodate future feature expansions.

Compliance: Adhere to university data protection policies and government regulations.

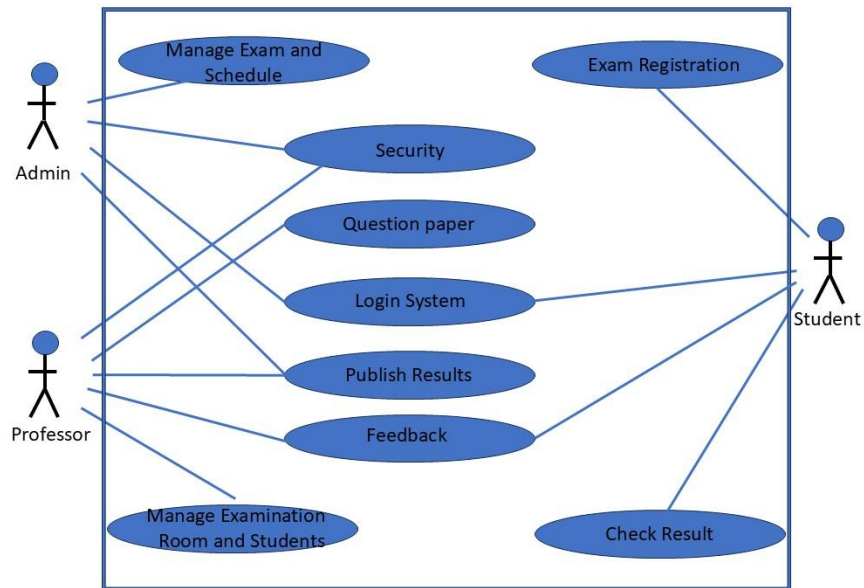
System Requirements

FOR PC	
OPERATING SYSTEM	Windows XP or Higher
PROCESSOR	500Mhz or faster
RAM	2gb or more
BROWSER	Google Chrome, Microsoft Excel or Brave
BROADBAND CONNECTION	500+ kbps
ADDITIONAL REQUIREMENTS	At least 1gb free disc space

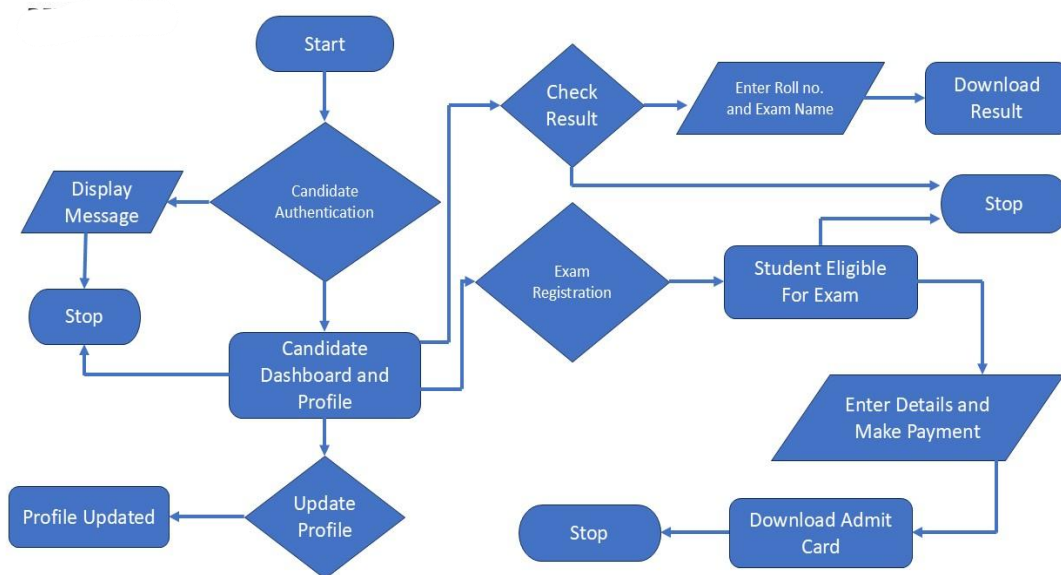
Context Diagram



Use Case Diagram



Sequence Diagram



Appendices

- Database schema diagrams
- API Documentation (if applicable)
- User Manual for Students and Administrators
- Glossary of terms related to exam registration